BME577: Drug Delivery **Spring 2025**
Ishaan Mehrotra
Dr. Fan Yuan

Project 1: Physiologically-Based Pharmacokinetic Analysis

1) Plot the concentration profiles of the drug in each organ as a function of time (from 0 to 100 min), using the baseline values of the model constants given in the Table.

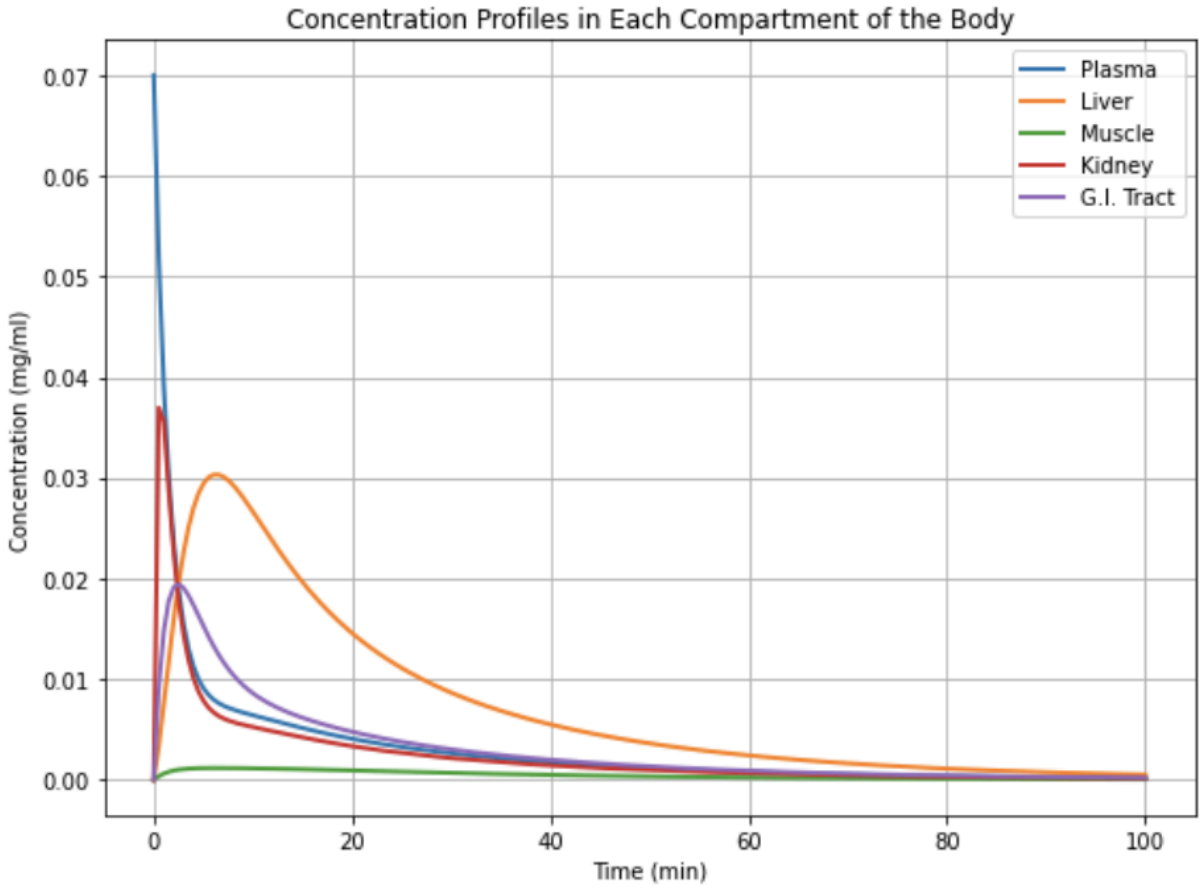


Figure 1. Time‑course of drug concentrations in Plasma, Liver, Muscle, Kidney, and G.I. Tract following an intravenous bolus dose of 3 mg/kg (for a 70 kg subject). The PBPK model was simulated over 100 minutes using the baseline parameter values.

At time zero, the drug is entirely in the plasma (at around 0.07 mg/ml), and its concentration there drops rapidly as the compound distributes into other tissues and undergoes elimination. The **liver** concentration rises quickly within the first few minutes, reflecting the organ's high blood flow and metabolic role, then declines as metabolism proceeds. The **kidney** also shows an early

peak before falling off due to excretion. The **G.I. tract** sees a moderate rise, reaching its maximum slightly later than the kidney, and then gradually declines. In contrast, the **muscle** concentration remains relatively low throughout, largely due to its small partition coefficient despite a large tissue volume. Overall, each compartment's concentration converges to very low values by about 60–100 minutes, illustrating the combined effects of distribution, metabolism, and renal excretion.

2) Determine how concentration profile changes if the dose of injection is increased by a factor of 7.

Figure 2. Time‑course comparison of drug concentrations in Plasma, Liver, Muscle, Kidney, and G.I. Tract under the baseline intravenous dose (3 mg/kg) and a 7‑fold higher dose. Peak

concentrations in all compartments increase substantially with the higher dose, while the overall shape of each time profile remains similar.

When the initial injection is increased seven‑fold, all compartments begin at substantially higher concentrations. The **plasma** still exhibits a rapid spike and subsequent decline, but its peak value scales up proportionally to the larger dose. In the **liver**, the higher dose leads to a more pronounced peak and a somewhat sustained mid‑range plateau before metabolism gradually lowers the concentration. The **kidney** shows a steep early peak reflecting the larger amount of drug being filtered and excreted. Meanwhile, the **muscle** remains comparatively lower in concentration—owing to its small partition coefficient—yet it follows a familiar rise‑and‑fall pattern at a higher absolute level. Finally, the **G.I. tract** sees a markedly higher early maximum, though it declines over time much like in the baseline case. Overall, while the shape of each profile is similar to the baseline scenario, the magnitudes are consistently amplified by the larger injection dose as expected.

3) Determine how the concentration profiles change if a) the drug is not metabolized in the body (i.e. $V_{max} = 0$) and b) the maximum metabolic rate is 7 times the baseline value.

Figure 3. Time‑course of drug concentrations in Plasma, Liver, Muscle, Kidney, and G.I. Tract under normal metabolism (solid lines) versus no hepatic metabolism (dashed lines, $V_{max} = 0$). In the absence of liver metabolism, drug clearance occurs solely via the kidney, resulting in higher peaks and slower overall elimination from each compartment.

With no liver metabolism available, the drug must be cleared entirely by the kidneys, causing concentrations in all compartments to remain higher for longer. In the **plasma**, the dashed line sits above the baseline curve throughout, reflecting slower overall elimination. The **liver** shows a taller and more prolonged peak because the usual enzymatic breakdown is absent. Similarly, the **kidney** faces a greater workload as it becomes the sole route of excretion, leading to a more sustained concentration. Both the **muscle** and **G.I. tract** also exhibit larger or longer‑lived peaks, as they lose the additional clearance normally provided by hepatic metabolism. Overall, the entire system displays a slower decline in drug levels compared to the baseline scenario.

Figure 4. Time‑course of drug concentrations in Plasma, Liver, Muscle, Kidney, and G.I. Tract under baseline metabolism (solid lines, V_{max} = 5 mg/min) versus a 7‑fold increase in metabolic capacity (dashed lines, V_{max} = 35 mg/min) The enhanced hepatic clearance accelerates the decline of drug levels in each compartment, leading to lower peaks and faster elimination relative to the baseline case.

With a seven-fold increase in the liver's metabolic capacity, the drug is cleared more rapidly from the bloodstream and distributed tissues. In **plasma**, the dashed curve falls below the original almost immediately, indicating accelerated overall clearance. The **liver** peaks quickly but drops faster than before, reflecting its heightened ability to metabolize the drug. In the **kidney**, the maximum concentration is somewhat reduced, as less drug survives in circulation to reach the renal compartment. Both the **muscle** and **G.I. tract** show smaller or shorter‑lived peaks compared to the baseline, again because of the intensified hepatic clearance. Ultimately, all compartments tend to lower concentrations sooner, underscoring the dominant role of liver metabolism when V_{max} is raised.

4) Determine how the drug clearance through the kidney affects the concentration profiles in each organ (i.e., changing the k_K by a factor of 20 in each direction).

Figure 5. Drug concentration‑time profiles in Plasma, Liver, Muscle, Kidney, and G.I. Tract for the baseline kidney clearance (solid lines) versus a 20‑fold reduction in clearance (dashed lines). The diminished renal excretion capacity ($k_K/20$) leads to higher and more prolonged drug levels across all compartments.

Reducing kidney clearance by a factor of 20 causes drug to linger in the system far longer than in the baseline scenario. In the **plasma**, the dashed line sits well above the original, reflecting slower elimination overall. The **kidney** itself shows a higher and more sustained peak, as it cannot excrete the drug efficiently. The **liver** still metabolizes some fraction, but a diminished renal outflow ultimately raises and prolongs concentrations in **muscle** and the **G.I. tract** as well. Consequently, every tissue experiences higher concentration levels for an extended duration.

Figure 6. Time‑course of drug concentrations in Plasma, Liver, Muscle, Kidney, and G.I. Tract for the baseline kidney clearance (solid lines) versus a 20‑fold increase in clearance (dashed lines). Enhanced renal excretion capacity sharply reduces and shortens the concentration profiles in every compartment, indicating faster systemic elimination.

With a **20‑fold increase** in kidney clearance, the drug is removed from circulation at a much faster rate, lowering concentrations across all tissues compared to the baseline. In **plasma**, the dashed line starts near the same initial point but drops more rapidly because of accelerated renal excretion. The **kidney** compartment itself sees a slightly earlier drop to very low levels, indicating more efficient clearance once the drug arrives there. In the **liver**, the concentration still initially rises but declines faster than in the baseline case due to the overall reduction in systemic availability. Similarly, **muscle** and **G.I. tract** both reach lower peaks and fall off sooner, as less drug remains in the bloodstream to partition into these compartments.